



1 Background & Problem

Learning centers often rely on fragmented tools for course management, enrollment, student follow-up, and parent communication. This creates inefficiency, weak visibility, and limited decision support. IQ PLUS was developed to provide one intelligent platform that connects all stakeholders and improves planning, monitoring, and academic decision-making.

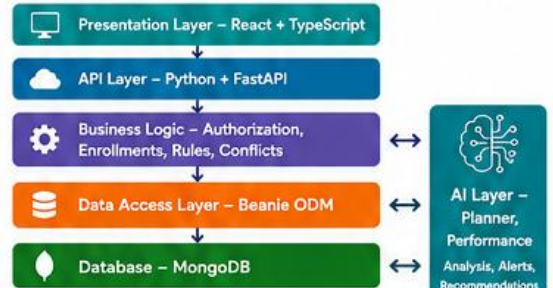
- Fragmented management workflows
- Limited visibility for parents and teachers
- Manual planning and conflict handling
- Need for academic insights and automation

2 Main System Requirements

- Multi-role platform: Admin, Teacher, Student, Parent
- Role-based dashboards and server-side RBAC
- Course management and enrollment workflows
- Academic planner with schedule conflict detection
- Performance analysis, alerts, and recommendations
- Materials, feedback, schedule, and chat support
- Bilingual-ready interface and AI outputs

3 Solution Overview

IQ PLUS is a full-stack smart learning management system that combines role-based workflows, layered architecture, and AI-assisted insights. The system centralizes academic data, supports daily operations, and helps users make better decisions.



4 Technologies & Development Tools

React	TypeScript
Python	FastAPI
MongoDB	Beanie ODM
REST APIs	Claude Code
Git / GitHub	Render / Vercel / MongoDB Atlas

5 Core Features

Multi-role dashboards	Course creation and enrollment	Parent-child academic tracking	Materials and PDF access
Chat and user search	Academic Planner	Performance score and insights	Alerts and action items

6 AI Capabilities

- Academic Planner: course analysis, workload estimation, conflict detection
- Performance Analysis: score derived from grades, attendance, teacher feedback, and progress trend
- Explain My Score: explains why a score is high or low
- AI Weekly Summary: role-specific weekly insights
- Action Items: practical next-step recommendations
- Multilingual AI outputs: Hebrew / English support

7 Engineering / Development Process



The project followed a structured engineering process including requirement analysis, layered design, iterative implementation, debugging, integration, role validation, and regression testing.

8 Project Metrics & Validation

4 Core Roles	204 / 204 Functional Tests Passed	Layered Architecture
AI-Enhanced Planning & Monitoring	RBAC Enforced on Server Side	Live Demo Ready

Testing focused on core flows, authorization, planner logic, dashboard loading, and stability under real user scenarios.

9 Challenges & Constraints

- Maintaining consistent data views across multiple roles
- Enforcing authorization beyond frontend visibility
- Handling planner conflicts and workload logic correctly
- Avoiding redundant API calls and N+1 query patterns
- Supporting multilingual AI outputs
- Deployment and cloud connectivity constraints

10 Results, Conclusions & Insights

IQ PLUS demonstrates how AI can enhance a learning management system without replacing core business rules. The final system improves visibility, planning, communication, and academic follow-up while preserving maintainability through layered design and data role boundaries.

- Strong end-to-end integration
- Better academic visibility for all stakeholders
- AI used as decision support, not as rule replacement
- Scalable foundation for future enhancements

11 Demo / Screens Preview #1 – Student Dashboard

Weekly Timetable
Click a class support

Sun	Mon	Wed	Thu
09:00 Algebra Fundamentals 09:00 – 10:00	09:00 מתמטיקה 09:00 – 10:00		
12:00 CleanHouse 12:00 – 13:00		12:00 CleanHouse 12:00 – 13:00	

7 Enrolled Courses **0%** Overall Score **4** AI Insights

My Courses

Course	Teacher	Action
MAT Algebra Fundamentals	David Cohen	Open →
MAT Advanced Calculus	David Cohen	Open →
ENG English Composition	Sarah Levy	Open →

Recent Chats [Go to Chats →](#)

- AD Advanced Calculus Ni everybody 2
- AL Algebra Fundamentals how are you sherbel?

Recent Updates [All Updates →](#)

- Great work! Your grade improved from 54.2 to 69.0 (NBS 2% increase)
- Your attendance has declined from 28.4% to 22.2%. Please prioritize attending classes.

12 Demo / Screens Preview #2 – Academic Planner

Academic Planner
AI-powered schedule planning and course recommendations

[Plan My Schedule](#) [Course Recommendations](#)

Select Courses

Course	Status	Time
Algebra Fundamentals MATH101	☐	Sun - 09:00-10:00
English Composition ENG101	☐	No schedule
Literature & Analysis ENG201	☒	No schedule
General Science SCI101	☐	No schedule
Physics Principles PHY101	☐	No schedule
Programming Fundamentals CS101	☐	No schedule
Chemistry Basics CHEM101	☐	No schedule
תכנות הנדסה 5 הנדסה	☐	Wed - 09:00-11:00

3 courses selected

Preferences

Preferred Days: ☒ Sun ☐ Mon ☐ Tue ☐ Wed ☐ Thu ☐ Fri

Preferred Free Day:

Max Courses: 5

Max Hours / Day: 6h

☐ Avoid early morning (before 9:00)

☐ Avoid late evening (after 20:00)

[Analyze Schedule](#)